

Sathyanarayan Rao, PhD

Research Scientist | Scientific R&D | Computational Modelling, Sensing & Scientific AI

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PROFILE

Computational scientist with more than fifteen years of research across plasma physics, optoelectronics, geoelectrical sensing of plant roots, agricultural digital twins and microwave remote sensing of soil and water. Across these domains, the common thread has been physics-based computation: modelling how signals interact with complex systems and inferring underlying processes from observations. Current work spans Earth-observation research at the Indian Institute of Science, scientific R&D as co-founder of Chakradhari Computational Technologies, and scientific evaluations that test where frontier AI systems reason soundly and where they fail. Research career across universities and national research centres in India, Europe and the United States.

Academic background: BE Electronics & Communication (VTU, India) · MS Electrical Engineering (UAH, USA) · MS Applied Physics (Alabama A&M, USA) · PhD Engineering Sciences (UCLouvain, Belgium)

AREAS OF EXPERTISE

- **Computational modelling.** Finite-element, particle-in-cell and Maxwell–Bloch simulation.
- **Electromagnetic & geoelectrical sensing.** Radar, microwave and ERT; forward and inverse problems.
- **Earth observation.** Soil moisture, watershed hydrology, decade-scale satellite time series.
- **Scientific AI.** Interpretable ML, frontier-model evaluation, causal and statistical inference.
- **Model coupling & digital twins.** Coupled crop–root models, interoperable scientific codes.
- **Research software & HPC.** Reproducible containers, parallel simulation, Python, C++ , Fortran.

PROFESSIONAL EXPERIENCE

Research Associate

Apr 2025 – Present

Department of Civil Engineering, Indian Institute of Science (IISc), Bengaluru

Hydro-REWARD (World Bank–supported): hydrological monitoring and decision support, Karnataka

- Independently develop and integrate satellite, field and computational analyses within the programme, from assembling datasets and building analysis frameworks through to publication.
- Microwave remote sensing of soil moisture: Sentinel-1 SAR retrieval, soil-texture-dependent and physically interpretable machine learning, and active–passive fusion with SMAP for continuous long-term records; radar calibration studies with CESBIO, France.
- Evaluation of watershed interventions in semi-arid catchments, combining decade-scale Sentinel-1 time series with quasi-experimental causal designs (difference-in-differences) and in-situ soil-moisture profiles.
- Curating multi-year, high-frequency soil hydrothermal observations from a monsoon catchment into a research dataset for model validation and reuse.

Co-Founder & Director

Oct 2025 – Present

Chakradhari Computational Technologies Private Limited, Bengaluru, India

Deep-tech scientific R&D company · cctech.in

- Co-founded a scientific R&D company; co-shape its scientific and technical direction across electromagnetic sensing, inverse problems, Earth observation, scientific AI and scientific software.
- Develop research partnerships with universities and research centres, including a formal MoU for joint research and funding proposals, to turn advanced scientific methods into usable computational systems.

Scientific AI Expert

May 2026 – Present

Mercor Intelligence

Remote, project-based · confidential frontier-AI research programmes

- Design scientific benchmarks and test environments used to evaluate and inform improvements to frontier AI systems across confidential research programmes.
- Build black-box inference problems in which a model must uncover a hidden physical or mathematical mechanism through its own choice of experiments, and simulation-based engineering tasks that require

designs meeting several interacting quantitative constraints.

- Write deterministic reference implementations, edge cases and grading logic; study model trajectories for shortcuts, leakage and brittle reasoning, and restructure problems to stay discriminating as capabilities advance.

Reviewer & Team Lead, Scientific AI Evaluation

Jan 2026 – Jul 2026

Turing

Project-based · confidential frontier-AI research programmes · task author, then reviewer and team lead

- Designed PhD-level problems in scientific computing, applied mathematics and multimodal STEM that demand derivation, computation and tool use rather than recall.
- Reviewed expert-authored evaluations for scientific correctness, reproducibility, difficulty and evaluation value, and decided on acceptance or rework; analysed repeated model runs to separate genuine reasoning failures from ambiguity, faulty ground truth or flawed benchmark design.
- Team lead for about 20 domain experts on a major scientific benchmark programme: led technical discussions, resolved disputed and rework cases, and supported consistent quality across the group.

Scientific Programmer

2023 – 2025

Forschungszentrum Jülich, Germany · PhenoRob Cluster of Excellence

- Computational modelling of agroecosystems within the PhenoRob digital-agriculture programme, linking process-based crop models with functional-structural root and plant models (CPlantBox).
- Designed coupling interfaces between independently developed scientific codes (C/C++, Fortran, Python), with time-step synchronisation and two-way exchange of state and parameters, including an architecture-driven root water uptake term for the AgroC crop model.
- Released public containerised environments for CPlantBox and DuMuX-ROSI, making coupled soil-root simulations reproducible on workstations and HPC clusters.

Research Associate

Sep 2022 – Aug 2023

Indian Institute of Science (IISc), Bengaluru

Independent Research & Development

2020 – 2022

India

Computational and machine-learning work in time-series modelling, signal analysis and applied data problems

Doctoral Researcher, F.R.S.-FNRS eRoot Project

2016 – 2020

UCLouvain, Belgium

- Developed coupled hydro-geophysical models of the soil-root system, using large-scale finite-element simulation on HPC (Jülich Supercomputing Centre) to predict electrical signatures of root architecture, anisotropy and root water uptake.
- Linked simulations to field and laboratory ERT and TDR measurements of plant response to drought; used machine learning to classify root systems from geophysical data.
- Visiting researcher, University of Bonn (2018), with Prof. Andreas Kemna, on geoelectrical imaging of roots.

Research Assistant

Aug 2015 – Mar 2016

Paderborn University, Germany · group of Prof. Torsten Meier

Computational optoelectronics: Fortran solvers for the Maxwell-Bloch equations of light-matter interaction in excitonic two-level systems (DFG-funded optical signal processing research).

Graduate Research Assistant, MS Research

2013 – 2014

Alabama A&M University, USA

Experimental applied physics: laser-induced photodegradation of dye molecules and photo-patterning of biosensing molecules on polymer thin films, characterised by atomic force microscopy.

Graduate Research Assistant

2010 – 2013

University of Alabama in Huntsville, USA

Computational plasma physics (NSF-funded): extended a particle-in-cell code from 2D electrostatic to 3D electromagnetic, parallelised with MPI/OpenMP, for helicon plasma and magnetic-nozzle studies; three papers in *Physics of Plasmas*.

PUBLICATIONS

Peer-reviewed journal articles

1. **Rao, S.** (2026). Temporal variability predicts learnability in repeated-entity spatiotemporal sensing. *Nordic Machine Intelligence*.
2. **Rao, S.**, Gupta, V., Murugesan, S., Zribi, M., & Muddu, S. (2026). An information proxy framework that explains why soil-specific radar calibration succeeds in sandy textures but fails in clay. *Remote Sensing Letters*, 17(4), 416–428. doi:10.1080/2150704X.2026.2647288
3. **Rao, S.**, Ahmadi, S. H., Leitner, D., Vianna, M., Bauer, F. M., Seidel, S. J., & Schnepf, A. (2026). Integrated Modeling Approaches for Agricultural Digital twins: the role of Process Based Models, Agent Based Models, Machine Learning, and Model Coupling. *in silico Plants*, 8(1), diag002. doi:10.1093/insilicoplants/diag002
4. Ehosioke, S., Nguyen, F., **Rao, S.**, Kremer, T., Placencia-Gomez, E., Huisman, J. A., Kemna, A., Javaux, M., & Garré, S. (2020). Sensing the electrical properties of roots: A review. *Vadose Zone Journal*, 19(1), e20082. doi:10.1002/vzj2.20082
5. **Rao, S.**, Lesparre, N., Flores-Orozco, A., Wagner, F., Kemna, A., & Javaux, M. (2020). Imaging plant responses to water deficit using electrical resistivity tomography. *Plant and Soil*, 454(1), 261–281. doi:10.1007/s11104-020-04653-7
6. **Rao, S.**, Meunier, F., Ehosioke, S., Lesparre, N., Kemna, A., Nguyen, F., Garré, S., & Javaux, M. (2019). Impact of Maize Roots on Soil–Root Electrical Conductivity: A Simulation Study. *Vadose Zone Journal*, 18(1), 190037. doi:10.2136/vzj2019.04.0037
7. Singh, N., **Rao, S.**, & Ranganath, P. (2013). Waves generated in the plasma plume of helicon magnetic nozzle. *Physics of Plasmas*, 20(3), 032111. doi:10.1063/1.4795734
8. Singh, N., & **Rao, S.** (2012). Plasma turbulence driven by transversely large-scale standing shear Alfvén waves. *Physics of Plasmas*, 19(12), 122303.
9. **Rao, S.**, & Singh, N. (2012). Numerical simulation of current-free double layers created in a helicon plasma device. *Physics of Plasmas*, 19(9), 093507. doi:10.1063/1.4754598

Peer-reviewed book chapters

1. **Rao, S.**, & Ranganath, P. (2025). Climate-Resilient Agriculture: Leveraging Language Models for Mitigation and Adaptation. In R. K. Srivastava & A. Chakraborty (Eds.), *Mitigation and Adaptation Strategies Against Climate Change in Natural Systems* (pp. 357–382). Springer, Cham. doi:10.1007/978-3-031-75968-0_19

Selected conference presentations

- Rao, S.**, et al. (2023). Random forest for soil moisture retrieval using satellite backscatter. Poster, Soil Moisture Workshop, IIT Bombay, India.
- Rao, S.**, et al. (2019). Investigation of electrical anisotropy as a root phenotyping parameter: numerical study with root water uptake. EGU General Assembly, Vienna, Austria. *Geophysical Research Abstracts*, 21.
- Rao, S.**, Meunier, F., Ehosioke, S., Lesparre, N., Nguyen, F., Kemna, A., Garré, S., & Javaux, M. (2017). A forward model for electrical conduction in soil–root continuum: a virtual rhizotron study. 4th International Workshop on Geoelectrical Monitoring (GELMON), Vienna, Austria.
- Rao, S.**, Ehosioke, S., Lesparre, N., Nguyen, F., & Javaux, M. (2017). Characterizing root system characteristics with Electrical resistivity Tomography: a virtual rhizotron simulation. EGU General Assembly, Vienna, Austria. *Geophysical Research Abstracts*, 19, EGU2017-4405.

Books and independent writing

- Rao, S.** (2026). *Three Days at Manikarnika: A Scientist Journey through Death and Life*. Kindle edition.
- Rao, S.** (2026). *How Not to Die in Indian Traffic: A Scientist's Survival Guide to Bengaluru Roads*. Pothi.com. ISBN 979-8903302376.
- Rao, S.** (2025). *Digital Twins: From Apollo Missions to Smart Farming*. Notion Press. ISBN 979-8900231457.

EDUCATION

PhD, Engineering Sciences, UCLouvain, Belgium

2020

Thesis: *Investigation of signatures of plant roots from non-invasive geo-electrical measurements*. Funded by the F.R.S.-FNRS eRoot project (T.1088.15).

MS, Applied Physics, Alabama A&M University, USA

2014

MS, Electrical Engineering, University of Alabama in Huntsville, USA

2012

BE, Electronics & Communication Engineering, Visvesvaraya Technological University, India

2010

METHODS & TOOLS

- **Scientific computing:** Python (NumPy, SciPy, pandas), C++, Fortran, MATLAB; finite-element and finite-difference methods; Gmsh; CPlantBox/CRootBox, DuMuX-ROSI, AgroC
- **Earth observation & sensing:** Sentinel-1 SAR, SMAP, Google Earth Engine, GDAL; ERT, TDR
- **Machine learning & statistics:** scikit-learn, PyTorch, TensorFlow; random forests, recurrent networks; difference-in-differences, BACI, event studies
- **Research software & HPC:** Docker, Linux, MPI/OpenMP, HPC batch systems
- **Agentic AI & scientific workflows:** CLI-driven, tool-using agent workflows (Claude Code, Codex) for scientific computing, benchmark construction, data curation and evaluation; Git/GitHub, containerised and cloud execution, reproducible evaluation pipelines

PROFESSIONAL CONTRIBUTIONS & SERVICE

- **Invited talks:** Sentinel-1 SAR and GPR for subsurface assessment, KCIRI, Coimbatore (Mar 2026); a computational scientist's view of quantum computing, Kumaraguru College of Technology (Oct 2025).
- **Peer review:** *Nordic Machine Intelligence*, *in silico Plants*, *PeerJ Computer Science*, *Plant and Soil*, *Vadose Zone Journal*, *Agricultural Research*, *Frontiers in Plant Science*.
- **Young Hydrologic Society:** Blog Committee member (Apr 2025–present), writing on computational hydrology, environmental modelling and digital twins.
- **Science communication:** creator of Compute Stories (YouTube, 2024–present), explaining computational modelling and scientific ideas.
- **Teaching:** Graduate teaching assistant for Physics 101 and 102, Alabama A&M University (2013–2014).